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Teo base schauder imp separable

Teorema 1. *Sea X un espacio de Banach, entonces*

$$\exists\,(e_n)_{n\in\mathbb{N}} \subset X \text{ base de Schauder} \implies X \text{ es separable.}$$

***Demostración*:** Suponemos que $\forall n \in \mathbb{N} : \|e_n\| = 1$ (si no, consideramos $\frac{e_n}{\|e_n\|}$). Sea $x \in X$, entonces $\exists!\,(a_n)_{n\in\mathbb{N}} \subset \mathbb{R} : \|x - \sum_{i=0}^{n} a_i e_i\| \xrightarrow[n\to\infty]{} 0$. Por tanto, dado $\varepsilon > 0$, $\exists N \in \mathbb{N} : \forall n \geqslant N : \|x - \sum_{i=0}^{n} a_i e_i\| < \frac{\varepsilon}{2}$.

Sea $(q_n)_{n\in\mathbb{N}} \subset \mathbb{K}$ tal que $\forall n \in \mathbb{N} : \Re(q_n), \Im(q_n) \in \mathbb{Q}$ y $|a_n - q_n| < \frac{\varepsilon}{2N}$. Entonces,

$$\begin{aligned}\left\| x - \sum_{i=0}^{N} q_i e_i \right\| &\leqslant \left\| x - \sum_{i=0}^{N} a_i e_i \right\| + \left\| \sum_{i=0}^{N} (a_i - q_i) e_i \right\| \\ &< \frac{\varepsilon}{2} + \sum_{i=0}^{N} |a_i - q_i|\, \|e_i\| < \frac{\varepsilon}{2} + N \cdot \frac{\varepsilon}{2N} = \varepsilon.\end{aligned}$$

Por tanto, el conjunto $\left\{ \sum_{i=0}^{n} q_i e_i : n \in \mathbb{N} \wedge \Re(q_i), \Im(q_i) \in \mathbb{Q} \right\}$ es denso en X y, como también es numerable, X es separable. ■

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